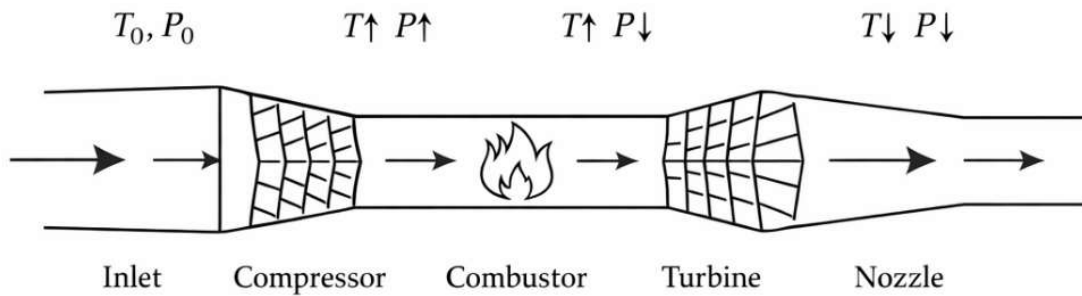


Turbojet Performance Optimization and Supersonic Inlet Design



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11 December 2025

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Nomenclature

Symbol	Description	Units
F	Thrust	[N]
F_{spec}	Specific thrust	[m/s]
$\dot{m}$	Mass flow rate	[kg/s]
u	Velocity	[m/s]
M	Mach number	[-]
P	Static Pressure	[kPa]
T	Temperature	[K]
W	Work	[J]
C_p	Specific heat at constant pressure	[J/(kg·K)]
γ	Ratio of specific heats	[-]
R	Gas constant for air	[J/(kg·K)]
P_0	Stagnation pressure	[Pa]
T_0	Stagnation temperature	[K]
π_c	Compressor pressure ratio	[-]
η	Efficiency	[-]
θ	Flow deflection angle	[Deg]
β	Shock angle	[Deg]

1. Introduction

A turbojet engine generates thrust by accelerating incoming air using combustion and a converging-diverging nozzle. This process is governed by compressible flow principles as well as thermodynamic cycle analysis. Figure 1.1 depicts the turbojet engine. It starts with air entering the compressor after going through the diffuser, then the compressed air leaving the compressor enters the combustor, where it is mixed with fuel and burned. Then it enters the turbine which powers the compressor. Then, in some situations, an afterburner is used to inject fuel after the turbine. Finally, the flow is accelerated in the converging-diverging nozzle, producing thrust.

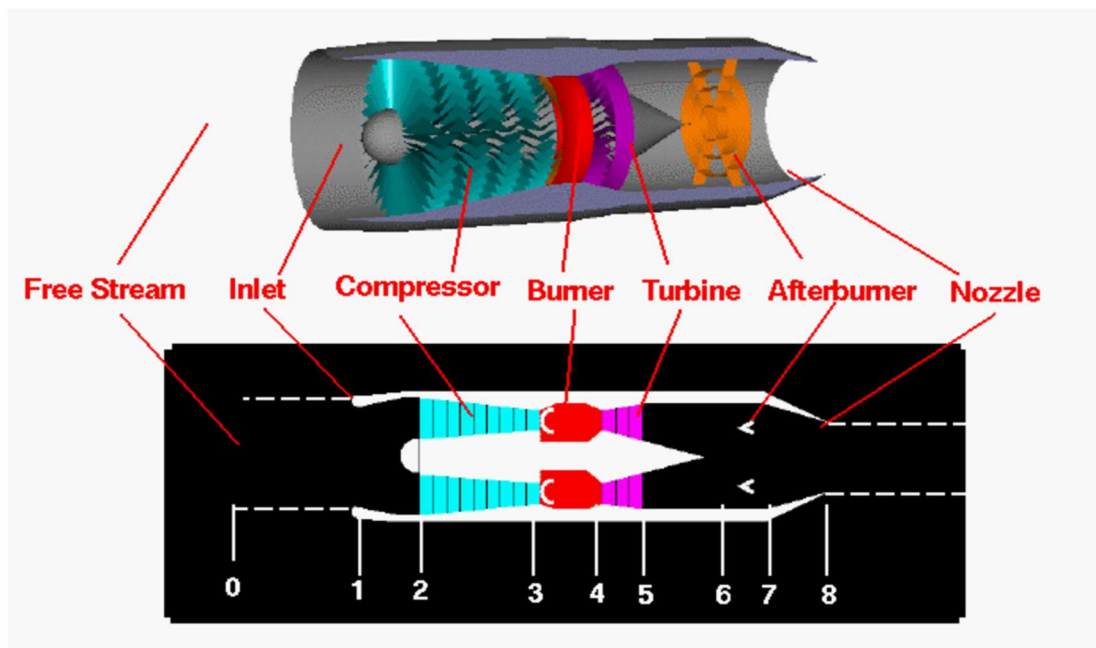


Figure 1.1. Turbojet diagram [1]

The thermodynamic behavior of the turbojet can be observed using the Brayton cycle, as shown in Figure 1.2. This diagram shows the relationship between temperature and entropy through each state of the process.

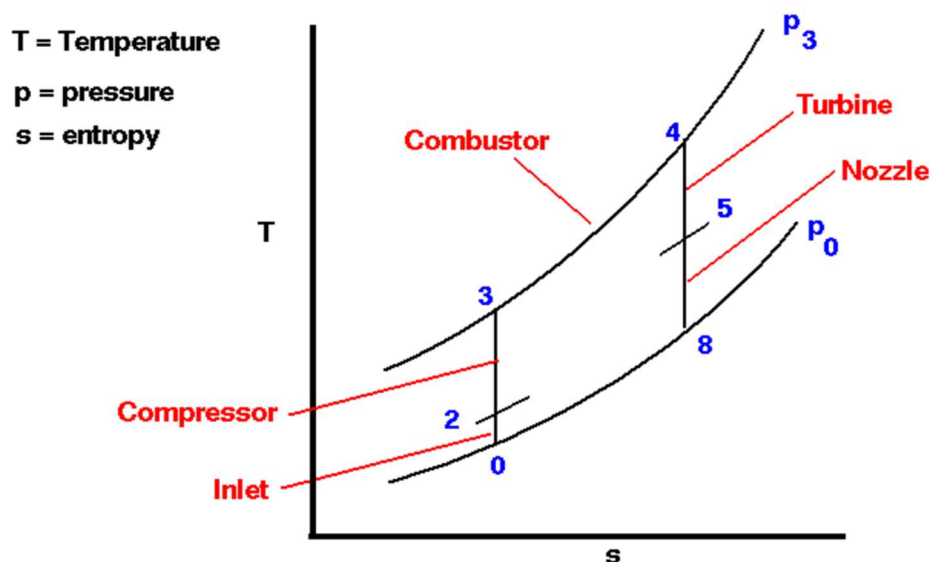


Figure 1.2. Brayton cycle diagram [2]

The first goal of this project is to determine the optimal compressor ratio to maximize specific thrust given certain conditions. Then, using this maximum specific thrust, calculate the mass flow rate required to produce 55kN of thrust and then the afterburner temperature required to increase the thrust to 70kN. Different ground and flight conditions will be studied for these results. The final goal of this project is to design a supersonic inlet that will decelerate the flow to subsonic speeds to ensure the engine works properly.

2. Methodology

This process describes how to calculate stagnation properties at each station of the turbojet engine using compressible flow equations and relations. Each stage assumes constant gas properties of air.

2.1. Diffuser (Stations i-1)

The first stage of the turbojet is the diffuser. This is where incoming air enters the engine. Equations (2.1.1-2.1.3) describe the inlet conditions of the engine. For ground operation, inlet Mach number is zero, but for supersonic flight, inlet conditions come from the isentropic deceleration caused by the diffuser.

$$M_i = M_\infty \quad (2.1.1)$$

$$T_i = T_\infty \quad (2.1.2)$$

$$P_i = P_\infty \quad (2.1.3)$$

$$u_i = M_i \sqrt{\gamma R T_i} \quad (2.1.4)$$

Equations (2.1.4) and (2.1.5) represent the isentropic deceleration. Note that for a Mach number of zero, stagnation properties leaving the diffuser and entering the compressor are the same as the ambient air properties.

$$T_{01} = T_{0i} = T_i \left(1 + \frac{\gamma - 1}{2} M_i^2 \right) \quad (2.1.4)$$

$$P_{01} = P_{0i} = P_i \left(1 + \frac{\gamma - 1}{2} M_i^2 \right)^{\left(\frac{\gamma}{\gamma - 1} \right)} \quad (2.1.5)$$

2.2 Compressor (Stations 1-2)

The next stage of the turbojet is the compressor. This is one of the most important components when it comes to thrust. The compressor raises the stagnation pressure and temperature before it enters the combustor. The compressor ratio in equation (2.2.1) explains how much each stagnation property increases, which is described in equation (2.2.2) and equation (2.2.3). For this project, the compressor is treated as isentropic.

$$\pi_c = \frac{P_{02}}{P_{01}} \quad (2.2.1)$$

$$P_{02} = \pi_c P_{01} \quad (2.2.2)$$

$$T_{02} = \pi_c^{\left(\frac{\gamma - 1}{\gamma} \right)} T_{01} \quad (2.2.3)$$

The compressor requires work to create this pressure change. In the ideal turbojet, the compressor is driven completely by the turbine, the work needed is given in equation (2.2.4).

$$W_c = C_p (T_{02} - T_{01}) \quad (2.2.4)$$

2.3 Combustor (Stations 2-3)

The combustor is where the high-pressure air exiting the compressor is mixed with fuel and burned to raise the stagnation temperature of the flow. Due to the deceleration from the compressor, the Mach number of the air in the combustor is very low, meaning the process can be treated as isobaric, yielding equation (2.3.1).

$$P_{03} = P_{02} \quad (2.3.1)$$

Equation (2.3.2) quantifies the change in stagnation temperature due to the combustor, which is reliant on many variables such as the mass flow rate of the air and fuel, the combustor efficiency, and the lower heating value of the fuel. However, in this project the exit temperature of the combustor is given at a set value and is not calculated using this equation.

$$T_{03} = T_{02} + \frac{\dot{m}_f \eta_b LHV}{\dot{m}_a C_p} \quad (2.3.2)$$

2.4 Turbine (Stations 3-4)

To power the compressor, the turbine extracts work from the combustion gases leaving the combustor. In this project, the turbine is assumed to operate isentropically. The turbine and the compressor operate on the same shaft, meaning the work done on the turbine by the air must be the same as the work done by the compressor, yielding equation (2.4.1).

$$W_T = C_p(T_{03} - T_{04}) = W_C \quad (2.4.1)$$

Using equation (2.4.1) with equation (2.2.4), equation (2.4.2) is derived which quantifies the stagnation temperature leaving the turbine. Equation (2.4.3) represents the change in stagnation pressure across the turbine using isentropic relations.

$$T_{04} = T_{03} - (T_{02} - T_{01}) \quad (2.4.2)$$

$$P_{04} = P_{03} \left(\frac{T_{04}}{T_{03}} \right)^{\left(\frac{\gamma}{\gamma-1} \right)} \quad (2.4.3)$$

2.5 C-D Nozzle (Stations 4-5)

The final section of the turbojet is the converging diverging (C-D) nozzle. This converts the thermal energy of the turbine exhaust into kinetic energy. This produces the high exit velocity that creates the thrust of the engine. Because the C-D nozzle operates isentropically, the stagnation properties across it do not change, yielding equations (2.5.1) and (2.5.2).

$$P_{05} = P_{04} \quad (2.5.1)$$

$$T_{05} = T_{04} \quad (2.5.2)$$

The ideal C-D nozzle is perfectly expanded, meaning that the exit pressure is the same as the ambient pressure, yielding equation (2.5.3).

$$P_5 = P_\infty \quad (2.5.3)$$

Using isentropic relations, equation (2.5.3) is derived to calculate the exit Mach number of the nozzle.

$$\frac{P_5}{P_{05}} = \left[1 + \frac{\gamma - 1}{2} M_5^2 \right]^{\left(\frac{\gamma}{\gamma - 1} \right)} \Rightarrow M_5 = \sqrt{\left[\left(\frac{P_5}{P_{05}} \right)^{\left(\frac{\gamma - 1}{\gamma} \right)} - 1 \right] \left[\frac{2}{\gamma - 1} \right]} \quad (2.5.3)$$

This exit Mach number can be used to calculate the exit temperature using isentropic relations as shown in equation (2.5.4).

$$\frac{T_5}{T_{05}} = \left[1 + \frac{\gamma - 1}{2} M_5^2 \right]^{-1} \Rightarrow T_5 = \frac{T_{05}}{1 + \frac{\gamma - 1}{2} M_5^2} \quad (2.5.4)$$

Equation (2.5.5) uses exit Mach number and temperature to calculate the exit velocity, a key parameter in determining thrust.

$$u_5 = M_5 \sqrt{\gamma R T_5} \quad (2.5.5)$$

Specific thrust is calculated using equation (2.5.7). At ground conditions inlet velocity (u_i) is zero, but in-flight conditions, this value comes from the flight Mach number as seen in equation (2.1.4).

$$\frac{F}{\dot{m}} = (u_5 - u_i) \quad (2.5.7)$$

Once specific thrust is known, equation (2.5.8) can be used to calculate the mass flow rate required to achieve a desired thrust.

$$\dot{m} = \frac{F}{(u_5 - u_i)} \quad (2.5.8)$$

2.6 Afterburner (Stations 4AB-5)

The afterburner is an auxiliary system that injects and burns additional fuel directly into the exhaust stream of the turbine. This process raises the stagnation temperature and increases the exit velocity of the C-D nozzle. Ideally, this is an isobaric process, meaning the stagnation pressure across the afterburner does not change. This assumption along with the assumption of a perfectly expanded C-D nozzle, means that the exit Mach number does not change, as the result of equation (2.5.3) would be the same. The stagnation temperature created by the afterburner is the same as the exit stagnation temperature of the C-D nozzle because the C-D nozzle is isentropic. Because of the above assumptions, the ratio of afterburner temperature and exit temperature are modeled by equation (2.6.1).

$$\frac{T_{0AB}}{T_5} = 1 + \frac{\gamma - 1}{2} M_5^2 \quad (2.6.1)$$

The required afterburner temperature for a desired thrust (F_{AB}) is calculated in equation (2.6.2). This uses equation (2.5.7) for thrust, equation (2.1.4) to calculate inlet velocity, equation (2.5.6) to calculate exit velocity, and the relationship described in (2.6.1).

$$T_{0AB} = \left[1 + \frac{\gamma - 1}{2} M_5^2 \right] \frac{\left(\frac{F_{AB}}{\dot{m}} + u_i \right)^2}{M_5^2 \gamma R} \quad (2.6.2)$$

2.7 Supersonic Inlet Design

At the high-speed flight condition of Mach 2.1, the air must be decelerated to a subsonic Mach number before it enters the compressor. To achieve subsonic air speed, a system of three oblique shocks followed by a normal shock is modeled. This is done by creating three wedge-like structures which cause oblique shocks that turn the flow inward

and reduce the Mach number of the incoming area. Then the normal shock decelerates the flow to sub-sonic conditions. The goal is to create a system that reduces the flow to sub-sonic conditions but minimizes the stagnation loss across the shocks. This is done by varying the angles of the wedges and finding the optimal combination of angles.

By using the oblique shock relations modeled in equations (2.7.1-2.7.5), the downstream Mach number and stagnation pressure loss is calculated for each oblique shock.

$$\tan \theta = 2 \cot \beta \left(\frac{M_1 \sin^2 \beta - 1}{M_1^2 (\gamma + \cos(2\beta)) + 2} \right) \quad (2.7.1)$$

$$M_{n1} = M_1 \sin \beta \quad (2.7.2)$$

$$M_{n2}^2 = \frac{1 + \frac{\gamma-1}{2} M_{n1}^2}{\gamma M_{n1}^2 - \frac{\gamma-1}{2}} \quad (2.7.3)$$

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)} \quad (2.7.4)$$

$$\frac{P_{02}}{P_{01}} = \left[\frac{(\gamma + 1) M_{n1}^2}{(\gamma - 1) M_{n1}^2 + 2} \right]^{\gamma/(\gamma-1)} \left[\frac{(\gamma + 1)}{2\gamma M_{n1}^2 - (\gamma - 1)} \right]^{1/(\gamma-1)} \quad (2.7.5)$$

After the third oblique shock, a normal shock decelerates the flow to subsonic speeds. The downstream Mach number and stagnation pressure loss due to the normal shock is calculated using equations (2.7.5) and (2.7.6).

$$M_2^2 = \frac{1 + \frac{\gamma-1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma-1}{2}} \quad (2.7.6)$$

$$\frac{P_{02}}{P_{01}} = \left[\frac{(\gamma + 1) M_1^2}{(\gamma - 1) M_1^2 + 2} \right]^{\gamma/(\gamma-1)} \left[\frac{(\gamma + 1)}{2\gamma M_1^2 - (\gamma - 1)} \right]^{1/(\gamma-1)} \quad (2.7.5)$$

3. Results and Discussion

By using the methodology described in Sections 2.1-2.6, the MATLAB code in appendix A is used to calculate the optimal compressor ratio for maximizing specific thrust for three different operating conditions. The code operates by iterating the process described for compressor ratios between 1 and 40. The code then graphs these compressor ratios as well as recording the specific thrust, airflow to achieve a thrust of 55kN, and the afterburner temperature required to increase thrust to 70kN for the optimal compressor ratio. The graph of compressor ratios is found in Figure 3.1, and the recorded data is found in Table 3.1.

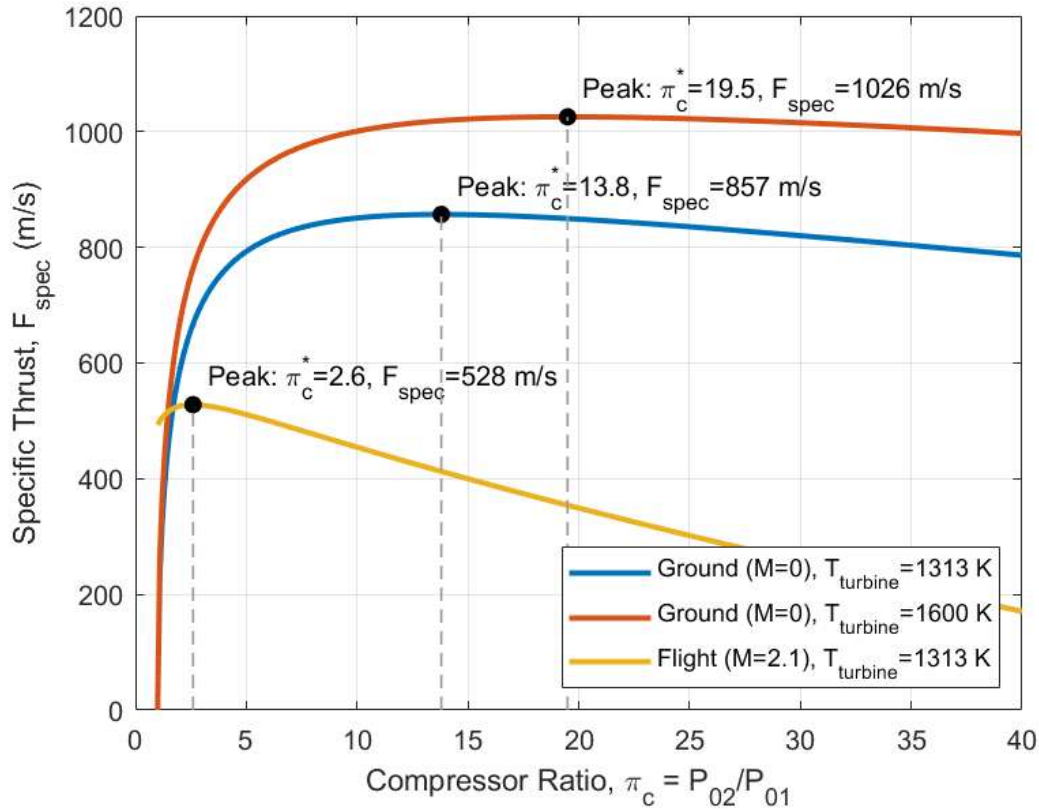


Figure 3.1. Specific Thrust vs Compressor Ratio

Table 3.1. Comparison of Turbojet Performance

	$M_\infty = 0$ Turbine Inlet: 1313K	$M_\infty = 0$ Turbine Inlet: 1600K	$M_\infty = 2.1$ Turbine Inlet: 1313K
Optimal Compressor Ratio	13.8	19.5	2.6
Maximum Specific Thrust [m/s]	856.91	1025.6	527.95
Airflow to Achieve Thrust=55kN [kg/s]	64.184	53.625	104.18
Afterburner Temp to Achieve Thrust=70kN [K]	1596.8	1957.5	1501.8

This data shows that increasing turbine inlet temperature raises the optimal compressor ratio and yields a higher maximum specific thrust. This makes sense as a more effective combustor adds more energy to the system which increases the work from the turbine as well as the enthalpy in the nozzle, providing more overall thrust.

However, when the engine operates at a high altitude and supersonic speeds, the optimal compressor ratio decreases along with the maximum specific thrust. The main contributor to this is the decrease in ambient pressure. At 15,000m, the pressure decreases from 100kPa to 12.11kPa. This pressure decreases along with the stagnation pressure loss from the inlet shocks severely decreases the amount of thrust the engine can produce.

To design an inlet that minimizes stagnation pressure loss, the MATLAB code in Appendix A uses the methodology described in Section 2.7 to calculate the total stagnation pressure loss across the oblique shocks and normal shock for

every combination of three angles ranging from 1-12 degrees. The code then saved the design with the highest stagnation pressure ratio. This combination of angles is 8.1 degrees, 8.9 degrees, and then 8.9 degrees again. The result of this can be found in Figures 3.2 and 3.3.

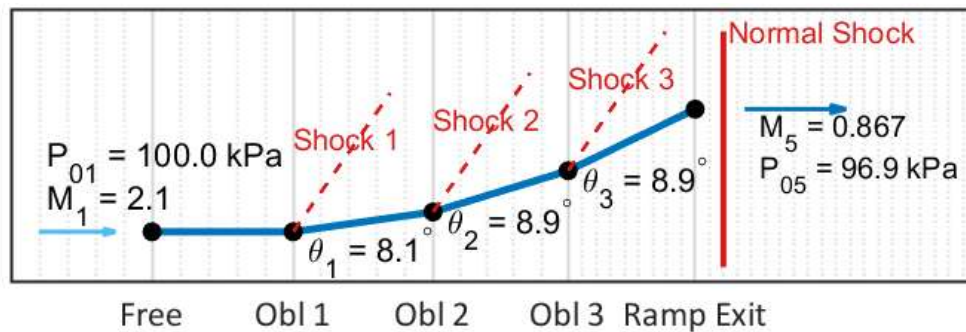


Figure 3.2. Optimal Supersonic Inlet Design

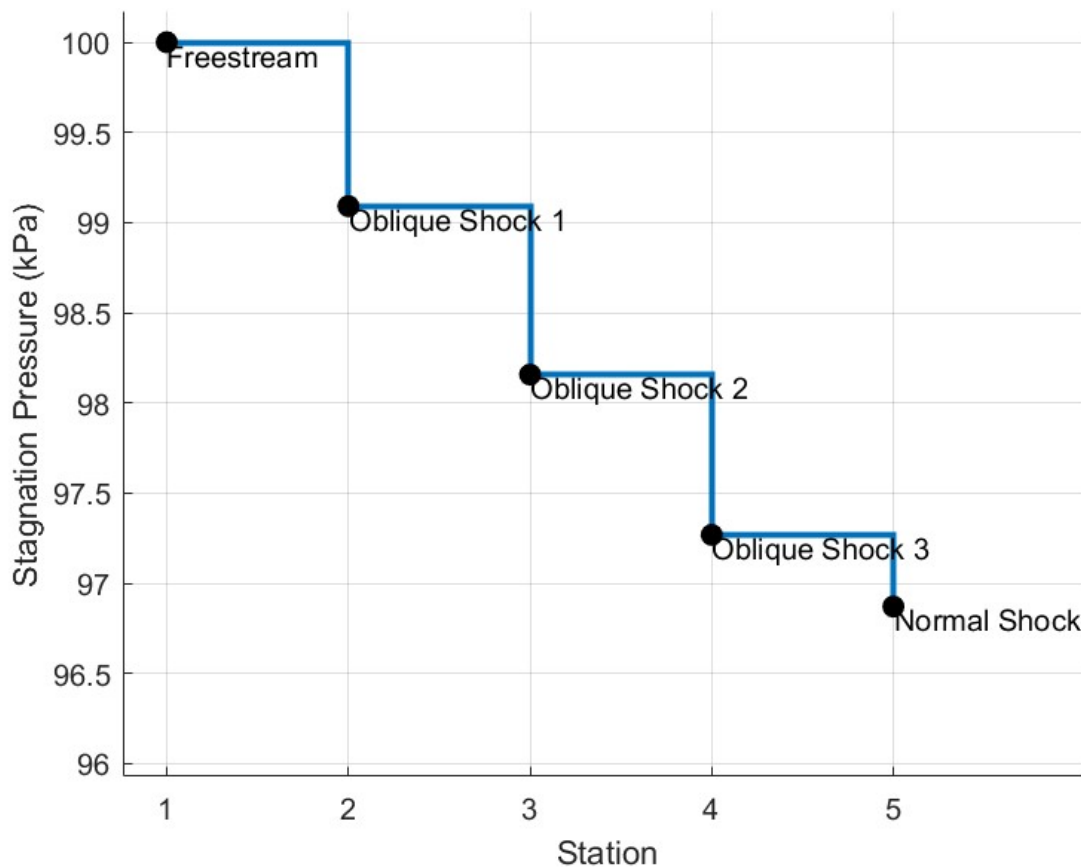


Figure 3.3. Stagnation pressure loss across inlet

These results show the loss in stagnation pressure with each shock. The oblique shocks greatly reduce stagnation pressure, and the normal shock slightly reduces stagnation pressure and brings the flow to subsonic speeds. This shows that evenly distributed angles preserve stagnation pressure better than one large angle.

4. Conclusion

By using the relations of compressible flow and the ideal Brayton cycle, this project succeeded in evaluating the performance of a turbojet engine given different ground conditions and flight conditions. The results concluded that a higher afterburner temperature yields higher thrust with a lower required airflow rate. When flying at supersonic speeds, thrust is significantly decreased and required airflow increases due to the lower ambient temperature and inlet shock losses.

The results also show that by raising the stagnation temperature of the flow entering the nozzle, thrust can be increased significantly. The supersonic inlet calculation revealed that using several oblique shocks followed by a normal shock can minimize the stagnation pressure loss as opposed to a single shock.

5. Appendix A: MATLAB code

```
%% Part 1

pi_c1=1:0.1:40; % Compressor Ratio guesses

Ue1=zeros([1 391]);

Me1=zeros([1 391]);

for i=1:1:391

    gamma=1.4; % constant

    R=287; % constant [J/kg K]

    % Station 1-2 Compressor (Isentropic)

    T01=293; % stagnation temperature station 1 [K]

    P01=100; % stagnation pressure station 1 [kPa]

    P02=pi_c1(i)*P01; % stagnation pressure station 2 [kPa]

    T02=T01*(pi_c1(i)^((gamma-1)/gamma)); % stagnation temperature station 2 [K]

    % Station 2-3 COmbustion Chamber (Constant P)

    T03=1313; % stagnation temperature station 3 [K]

    P03=P02; % stagnation pressure station 3 [kPa]

    % Station 3-4 Turbine (Isentropic)

    T04=T03-T02+T01; % stagnation temperature station 4 [K]

    P04=P03*((T04/T03)^(gamma/(gamma-1))); % stagnation pressure station 4 [kPa]

    % Station 4-e C-D Nozzle

    Pe=100; % pressure at exit [kPa]

    Me1(i)=sqrt((((Pe/P04)^(-(gamma-1)/gamma))-1)*(2/(gamma-1))); % Mach number at exit

    Te=T04/(1+((gamma-1)/2)*(Me1(i)^2)); % stagnation temperature at exit [K]

    Ue1(i)=Me1(i)*sqrt(gamma*R*Te); % velocity at exit (Specific Thrust) [m/s]
```

end

[Max_Spec_Thrust_part1,index1]=max(Ue1); % max specific thrust [m/s]

Optimal_Comp_Ratio_part1=pi_c1(index1); % compressor ratio yielding max specific thrust

T=55000; % desired thrust [N]

m_dot1=T/Max_Spec_Thrust_part1; % air mass flow rate [kg/s]

% Afterburner Temp:

F_AB=70000; % desired thrust [N]

Me=Me1(index1); % Exit mach Number

mdot=m_dot1; % air flow [kg/s]

T_AB1=((F_AB/mdot)^2)/(gamma*R*Me^2))*(1+((gamma-1)/2)*Me^2); % afterburner temperature to achieve thrust

%% Part 2

pi_c2=1:0.1:40; % Compressor Ratio guesses

Ue2=zeros([1 391]);

Me2=zeros([1 391]);

for i=1:1:391

gamma=1.4; % constant

R=287; % constant [J/kg K]

% Station 1-2 Compressor (Isentropic)

T01=293; % stagnation temperature station 1 [K]

P01=100; % stagnation pressure station 1 [kPa]

P02=pi_c2(i)*P01; % stagnation pressure station 2 [kPa]

T02=T01*(pi_c2(i)^((gamma-1)/gamma)); % stagnation temperature station 2 [K]

```

% Station 2-3 COmbustion Chamber (Constant P)

T03=1600; % stagnation temperature station 3 [K]

P03=P02; % stagnation pressure station 3 [kPa]


% Station 3-4 Turbine (Isentropic)

T04=T03-T02+T01; % stagnation temperature station 4 [K]

P04=P03*((T04/T03)^(gamma/(gamma-1))); % stagnation pressure station 4 [kPa]


% Station 4-e C-D Nozzle

Pe=100; % pressure at exit [kPa]

Me2(i)=sqrt((((Pe/P04)^(-(gamma-1)/gamma))-1)*(2/(gamma-1))); % Mach number at exit

Te=T04/(1+((gamma-1)/2)*(Me2(i)^2)); % stagnation temperature at exit [K]

Ue2(i)=Me2(i)*sqrt(gamma*R*Te); % velocity at exit (Specific Thrust) [m/s]

end


[Max_Spec_Thrust_part2,index2]=max(Ue2); % max specific thrust [m/s]

Optimal_Comp_Ratio_part2=pi_c2(index2); % compressor ratio yielding max specific thrust

T=55000; % desired thrust [N]

m_dot2=T/Max_Spec_Thrust_part2; % air mass flow rate [kg/s]


% Afterburner Temp:

F_AB=70000; % desired thrust [N]

Me=Me2(index2); % Exit mach Number

mdot=m_dot2; % air flow [kg/s]


T_AB2=((F_AB/mdot)^2)/(gamma*R*Me^2)*(1+((gamma-1)/2)*Me^2); % afterburner temperature to achieve thrust

%% Part 3

```

```

pi_c3=1:0.1:40; % Compressor Ratio guesses

F3=zeros([1 391]);
Me3=zeros([1 391]);

for i=1:1:391

    Mi=2.1; % inlet mach number

    gamma=1.4; % constant

    R=287; % constant [J/kg K]

    % Station i-1 Diffuser

    Ti=216; % inlet temperature [K]

    Pi=12.11; % inlet pressure [kPa]

    Ui=Mi*sqrt(gamma*R*Ti); % inlet air velocity [m/s]

    T0i=Ti*(1+((gamma-1)/2)*(Mi^2)); % inlet stagnation temperature [K]

    T01=T0i; % station 1 stagnation temperature [K]

    P0i=Pi*((1+((gamma-1)/2)*(Mi^2))^(gamma/(gamma-1))); % inlet stagnation pressure [kPa]

    P01=P0i; % stagnation pressure at station 1 [kPa]

    % Station 1-2 Compressor (Isentropic)

    P02=pi_c3(i)*P01; % stagnation pressure at station 2 [kPa]

    T02=T01*(pi_c3(i)^((gamma-1)/gamma)); % stagnation temperature at station 2 [K]

    % Station 2-3 Combustion Chamber (Constant P)

    T03=1313; % stagnation temperature at station 3 [K]

    P03=P02; % stagnation pressure at station 3 [kPa]

    % Station 3-4 Turbine (Isentropic)

    T04=T03-T02+T01; % stagnation temperature station 4 [K]

    P04=P03*((T04/T03)^(gamma/(gamma-1))); % stagnation pressure station 4 [kPa]

```

```

% Station 4-e C-D Nozzle

Pe=12.11; % pressure at exit [kPa]

Me3(i)=sqrt((((Pe/P04)^(-(gamma-1)/gamma))-1)*(2/(gamma-1))); % Mach number at exit
Te=T04/(1+((gamma-1)/2)*(Me3(i)^2)); % stagnation temperature at exit [K]
Ue3=Me3(i)*sqrt(gamma*R*Te); % velocity at exit (Specific Thrust) [m/s]
F3(i)=Ue3-Ui;

end

[Max_Spec_Thrust_part3,index3]=max(F3); % max specific thrust [m/s]
Optimal_Comp_Ratio_part3=pi_c3(index3); % compressor ratio yielding max specific thrust
T=55000; % desired thrust [N]
m_dot3=T/Max_Spec_Thrust_part3; % air mass flow rate [kg/s]

% Afterburner Temp:
F_AB=70000; % desired thrust [N]
Me=Me3(index3); % Exit mach Number
mdot=m_dot3; % air flow [kg/s]

T_AB3=((F_AB/mdot+Ui)^2)/(gamma*R*Me^2)*(1+((gamma-1)/2)*Me^2); % afterburner temperature to
achieve thrust

%% Part 4
gamma=1.4;
M1=2.1;

max_P05_P01=0;
n=40;
theta=linspace(1,12,n);
for x=1:1:n % angle 1
    for y=1:1:n % angle 2

```



```

for z=1:1:n % angle 3

% Oblique 1:
[M2,P02_P01]=Oblique_Shock(M1,theta(x),gamma);

% Oblique 2:
[M3,P03_P02]=Oblique_Shock(M2,theta(y),gamma);

% Oblique 3:
[M4,P04_P03]=Oblique_Shock(M3,theta(z),gamma);

% Normal Shock
[M5,P05_P04]=Normal_Shock(M4,gamma);

% Stagnation pressure ratio
P05_P01=P02_P01*P03_P02*P04_P03*P05_P04;

% Store data for optimal angles
if P05_P01 > max_P05_P01
    max_P05_P01=P05_P01;
    theta1=theta(x);
    theta2=theta(y);
    theta3=theta(z);

M5_optimal=M5;

P05_P04_optimal=P05_P04;
P04_P03_optimal=P04_P03;
P03_P02_optimal=P03_P02;
P02_P01_optimal=P02_P01;

```

```
        end
    end
end
end
```

```
angles=[theta1 theta2 theta3]
```

```
P01=100; % kPa
P02=P01*P02_P01_optimal; % kPa
P03=P02*P03_P02_optimal; % kPa
P04=P03*P04_P03_optimal; % kPa
P05=P04*P05_P04_optimal; % kPa
```

6. References

[1] NASA Glenn Research Center, "Turbojet Engine Schematic."

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[2] NASA Glenn Research Center, "The Brayton Cycle."

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